MILITARY SPECIFICATION

POWER SUPPLY, TELETYPEWRITER, LOW LEVEL

1. SCOPE

1.1 This specification covers requirements for a power unit capable of delivering regulated dc for low level (±6 V) teletypewriter signal circuits.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

- MIL-S-901 - Shock Tests, H.I.(High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for
- MIL-Q-9858 - Quality Program Requirements
- MIL-M-15071 - Manuals, Equipment, and Systems
- MIL-E-16400 - Electronic Equipment, Naval Ship and Shore: General Specification
- MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of
- MIL-R-22732 - Reliability Requirements for Shipboard and Ground Electronic Equipment

STANDARDS

MILITARY

- MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
- MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
- MIL-STD-189 - Racks, Electrical Equipment, 19-Inch and Associated Panels
- MIL-STD-454 - Standard General Requirements for Electronic Equipment
- MIL-STD-461 - Electromagnetic Interference Characteristics, Requirements for Equipment
- MIL-STD-701 - Lists of Standard Semiconductor Devices
- MIL-STD-781 - Reliability Tests: Exponential Distribution
- MIL-STD-785 - Reliability Program for Systems and Equipments, Development and Production

Source: https://assist.dla.mil -- Downloaded: 20261O-20T13:49Z
Check the source to verify that this is the current version before use.

PUBLICATIONS

MILITARY

MIL-HDBK-217 - Reliability Stress and Failure Rate Data for Electronic Equipment

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements.- The equipment covered by this specification shall meet the basic design objectives of and conform to the requirements of MIL-E-16400 to the extent specified herein.

3.2 First article sample.- Prior to beginning production, a sample shall be tested as specified in 4.3 (see 6.3).

3.3 Modular construction provisions of MIL-E-16400 are not applicable.

3.4 Design requirements.-

3.4.1 General design.- The power supply shall consist basically of two identical units, each capable of producing a positive 6-volt, 5 ampere regulated dc output, and a negative 6-volt, 5 ampere regulated dc output from a 120-volt signle phase, 50 to 60 Hz source. Solid state rectifiers with regulating and control circuits shall be used to achieve the performance specified in 3.4.4.

3.4.2 Material and parts.- The material and parts used in the construction of this equipment shall be selected in accordance with MIL-E-16400. Rectifier and regulator elements shall be solid state selected in accordance with MIL-STD-701. Nonstandard parts, if necessary to meet the required performance must be approved in accordance with MIL-E-16400. Items or processes which require approval of the procuring activity are listed in MIL-E-16400.

3.4.3 Environmental conditions.-

3.4.3.1 Temperature.- The operating temperature range shall be class 4 in accordance with MIL-E-16400.

3.4.3.2 Humidity.- The humidity range shall be in accordance with MIL-E-16400.

3.4.3.3 Thermal design.- Thermal design shall be in accordance with MIL-E-16400.

3.4.4 Performance.-

Source: https://assist.dla.mil -- Downloaded: 20261O-20T13:49Z
Check the source to verify that this is the current version before use.

3.4.4.1 Primary power supply line voltage and frequency.- Primary power will be single phase 120 V ac ±10 percent at 50 or 60 Hz ±5 percent.

3.4.4.2 Output characteristics.-

3.4.4.2.1 Voltage and current.- Rated output shall be 5 amperes. Output voltage shall be adjustable between 5 and 7 volts on either side of the neutral output terminal. The output circuit shall be ungrounded.

3.4.4.3 Regulation.- Regulation shall be maintained within ± 1 percent on either polarity and for any setting between 5 and 7 volts for the full range of primary power regulation and zero to full rated output.

3.4.4.4 Ripple.- Ripple shall be less than 10 mV RMS for any load within the rated output.

3.4.4.5 Parallel operation.- The units shall be designed for parallel operation. Provisions for parallel or separate output option shall be by convenient strapping within the units. Either of the pair of units comprising the power supply shall be removable without interrupting or impairing the operation of the remaining unit.

3.4.4.6 Overload protection.- The output current shall be limited (non-adjustable) to the full rated output for operation under any overload condition including a direct short circuit, regardless of how long maintained. Supply shall automatically restore to normal operation upon overload removal.

3.4.4.7 Meters.- Each of the two units comprising the power supply shall have a voltmeter and an ammeter switchable to indicate either positive or negative output. The voltmeters shall be suppressed zero, expanded scale type with scale length at least 1.85 inches and 2 percent full scale accuracy, indicating 4 to 8 volts dc. The ammeters shall read zero to 5 amps dc, with same scale length and accuracy.

3.4.4.8 Power switch.- Each unit shall have a power switch to disconnect the ac imput in the "off" position. A power-on indicator light with a red jewel shall be mounted on the panel.

3.4.4.9 Fuses.- There shall be a fuse, with blown fuse indicator in each lead of the imput circuits.

3.4.4.10 Connection.- Connection for ac power, positive and negative dc output, battery ground and chassis ground shall be made from screw type terminals at the rear of the chassis. Provision for cable entrance shall be made at the rear.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:49Z
Check the source to verify that this is the current version before use.

3.4.5 Mechanical design.- Mechancial design shall be in accordance with MIL-E-16400.

3.4.5.1 Enclosure.- The equipment shall employ a cabinent type construction basically designed for top of table or shelf mounting, but shall also be suitable for relay rack mounting in accordance with MIL-STD-189 upon attachment of rack mounting brackets. Degree of enclosure shall be open protected in accordance with MIL-STD-108.

3.4.5.2 Mounting brackets.- Rack mounting brackets shall be secured to the sides of the cabinet by means of screws or bolts which thread into reinforced members attached inside the walls of the cabinet. The mounting brackets and cabinet shall be constructed so as to preclude the need for additional support when mounted in a standard rack.

3.4.5.3 Front panel.- All controls, fuses, indicators and switches shall be mounted on the front panel. Screw driver adjustment shall be provided for voltages specified in 3.4.4.2.

3.4.5.3.1 Panel layout shall be submitted for approval in accordance with MIL-E-16400.

3.4.5.4 Shock.- The equipment shall be designed to withstand the shock test grade A, class I, type C in accordance with MIL-S-901.

3.4.5.5 Vibration and inclination.- The equipment shall be subject to the vibration and inclination tests of MIL-E-16400.

3.4.6 Size and weight.- Width shall be such as to permit mounting in accordance with 3.4.5.1. Height shall be standard 7 inches. Depth shall not exceed 9 inches. Weight shall be minimum to achieve the specified performance, reliability and maintainability and not to exceed 50 pounds.

3.4.7 Painting.- Painting and treatments shall be in accordance with MIL-E-16400.

3.4.8 Test points.- Significant voltages, currents, and wave forms shall be measurable at test points to facilitate alignment and determination of circuit faults, in accordance with MIL-E-16400.

3.5 Radio interferences and undesired radiation.- The equipment shall meet the radio interference and undesired radiation requirements of MIL-STD-461 for class IC equipments.

3.6 Reliability

3.6.1 Quantitive reliability requirement.- The specified MTBF (θ_o as defined by MIL-STD-781) of this equipment is 10,000 hours.

Source: https://assist.dla.mil -- Downloaded: 20261O-20T13:49Z
Check the source to verify that this is the currentversion before use.

3.6.2 Reliability program plan.- The contractor shall submit a reliability program plan in accordance with and in paragraph sequence of MIL-STD-785. Particular attention should be paid to those sections of MIL-STD-785 which require submission of detailed tasks, procedures to implement and control these tasks and time schedule. The proposed reliability program plan shall be submitted as a separate and complete entity within the total system or equipment/project delineation. The reliability program plan as approved and incorporated into the contract becomes the basis for contractual compliance. The plan, including reliability test plans, shall describe an intregated effort within the total project; it shall provide information to show how the contractor will meet specified quantitative reliability requirements during development and manufacture. The final reliability program plan shall be submitted for Government approval.

3.6.3 Reliability prediction.- The contractor shall submit to the procuring activity a reliability prediction in accordance with "How to make a reliability prediction based on stress factors and part populations" as shown in MIL-HDBK-217. The predicition shall assume a 50°C ambient temperature. The contractor shall evaluate the predication with respect to the reliability expected from a production unit.

3.6.4 Source of failure rates.- The primary source of failure rate data shall be MIL-HDBK-217. For items not covered by MIL-HDBK-217, other sources of data may be used subject to the approval of the procuring activity. When such alternate sources of data are proposed, adequate information shall be submitted to enable the procuring activity to evaluate the validity and applicability of the results derived therefrom.

3.6.5 Reliability reports.- Reports providing a complete accounting of progress on each element defined by the reliability program plan, results achieved and status of actions to resolve problems shall be submitted monthly to the procuring activity.

3.7 Supplementary requirements.- Supplementary requirements of MIL-E-16400 relating to serial numbers, parts identification, Government nomenclature, identification plates, and markings shall apply to this equipment.

3.8 Manuals.- When required by contract or order, instruction manuals shall be type I in accordance with MIL-M-15071.

3.9 Safety.- Safety shall be in accordance with Requirement 1 of MIL-STD-454.

3.10 Workmanship.- Workmanship shall be in accordance with Requirement 9 of MIL-STD-454.

Source: https://assist.dla.mil -- Downloaded: 20261O-20T13:49Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor's quality assurance system and qualtiy program plan. - The contractor shall provide and maintain an effective inspection and quality control system acceptable to the Government covering the supplies under the contract. The quality control systems shall be in accordance with MIL-Q-9858. A quality program plan, as it would apply to the requirements of this specification and contract, shall be submitted to the cognizant Government inspector, for evaluation of acceptability prior to first article inspection. (An advance copy shall be forwarded to the command or agency concerned). If first article inspection is not required, the quality program plan shall be submitted. The written description shall include flow charts showing the complete flow of material from inspection on receipt, through all manufacturing processes, to final shipment. The various processes and inspection positions shall be clearly identified on the flow charts and cross reference to the detailed inspection procedures and criteria of the written description of the quality control system. Any changes to the quality control system or the quality program plan shall be submitted to the command or agency concerned for approval prior to use.

4.1.2 Government verification.- All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of (a) surveillance of the operations to determine that practice, methods, and procedures of the written system description are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to assure compliance with all the requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.3 Failure reporting.- Failure reporting shall be in accordance with MIL-E-16400.

4.1.4 Test, test equipment, and test procedures.- The contractor shall submit a list of proposed tests and a copy of the written test procedures, including test equipment and tolerance limits to be used in testing first

Source: https://assist.dla.mil -- Downloaded: 202610-20T13:49Z
Check the source to verify that this is the currentversion before use.

article equipment and quality conformance inspection the the command or agency concerned, via the Government inspector, for review and approval, prior to beginning first article tests. An advance copy shall be forwarded to the command or agency concerned. Each test shall be identified in accordance with 4.2 as to classification. The contractor may, at his option, submit the production portion prior to beginning production. Test shall not begin until approval is received.

4.2 Classification of inspection.- The method of examination and testing the equipment fall within the following classifications:

(a) First article inspection
(b) Quality conformance inspection

(1) Production inspection (see 4.4.1)
(2) Production control inspection (see 4.4.2)
(3) Environmental inspection (see 4.4.3)
(4) Inspection of preparation for delivery (see 4.5.4)

4.3 First article inspection.- Unless otherwise specified (see 6.3), one unit shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in table I.

Table I. First article inspection

Inspection	Reference
Surface examination	4.5.1
Operating test	4.5.2
Weights and dimensions	MIL-E-16400
Supply line voltage and frequency	MIL-E-16400
Heat test	MIL-E-16400
Temperature tests	MIL-E-16400
Humidity test (see 4.3.1)	MIL-E-16400
Power	MIL-E-16400
Controls and control circuits	MIL-E-16400
Accelerated life test	MIL-E-16400
Shock (see 3.4.5.4)	MIL-S-901
Vibration and inclination	MIL-E-16400
Reliability	Test level No. 2, MIL-R-22732 shall be used for determining environment for reliability testing
Electromagnetic interference characteristics	MIL-STD-461, class IC

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:49Z
Check the source to verify that this is the current version before use.

4.3.1 Humidity tests required herein shall be conducted to verify compliance with the detailed requirements specified in MIL-E-16400.

4.3.2 First article inspection report.- The first article inspection report shall be submitted to the command or agency concerned, via the Government inspector, for review and approval. An advance copy shall be forwarded to the command or agency concerned. The first article inspection report shall contain results of all tests and shall include completed Form NAVEXOS 3373 for shock test. First article approval will not be granted unless the report is complete. Any production entered into prior to approval of the first articles inspection report will be at the contractor's risk.

4.3.3 First article model acceptance.- Acceptance of the first article model shall be contingent upon approval by the command or agency concerned with the first article inspection report. Any further laboratory testing of the first model may be made at the discretion of the Government.

4.4 Quality conformance inspection.-

4.4.1 Production inspection.- Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process, such as functional and performance tests at a limited number of points, tests which detect deviations from designs, tests of adjustment, and tests which detect hidden defects of material. Production inspection shall consist of the tests approved in accordance with 4.1.4 and shall include the following:

(a) Surface examination (4.5.1)
(b) Operating test (4.5.2)

4.4.2 Production control inspection.- Production control inspection, including sampling, shall conform to table II and to the procedures for special inspection levels of MIL-STD-105. The inspection level shall be S-3 for normal, tightened, and reduced inspection. Production control inspection shall be performed on equipment that has passed production inspection. The equipment shall satisfactorily meet the requirements of production control inspection prior to release for shipment.

Table II. Production control inspection

Inspection	Reference	AQL percent
Weights and dimensions	MIL-E-16400	6.5
Supply line voltage and frequency	MIL-E-16400	6.5
Heat test	MIL-E-16400	6.5
Power	MIL-E-16400	6.5
Electromagnetic interference	MIL-STD-461	6.5
Controls and control circuits	MIL-E-16400	6.5

Source: https://assist.dla.mil -- Downloaded: 202640-20T13:49Z
Check the source to verify that this is the currentversion before use.

4.4.2.1 Rejected lots.- If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor may also rework a rejected lot to correct the defective units and reinspect the lot, using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.3 Environmental inspection.- Environmental inspection shall be conducted to prove the durability of the equipment to withstand the effects of extremes of environment, service conditions, and hard usage. Environmental inspection shall be performed on equipment or equipments that have passed production inspection in accordance with 4.4.1 and shall include the tests listed in table III.

4.4.3.1 Environmental inspection samples.- Samples for the environmental inspection shall be selected without regard to their quality in accordance with the following:

(a) One sample from the first 50 production units fabricated
(b) One sample from each successive 100 units produced or fraction thereof

Table III. Environmental inspection

Inspection	Reference
Temperature	MIL-E-16400
Humidity (see 4.3.1)	MIL-E-16400

4.4.4 Nonconformance and retest. - If a sample unit fails a test specified in 4.4.2 or 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken to correct units of product which were manufactured with the same conditions, materials, and processes. The equipment shall then be retested.

4.4.5 Reinspection of conforming production control and environmental test units.- Unless otherwise specified, sample units which have been subjected to and passed production control and environmental inspection may be accepted on the contract provided they are resubjected to and pass production inspection after repair of all visible damage.

4.4.6 Quality conformance inspection report.- The contractor shall submit to the Government inspecting activity prior to shipment of the respective equipment a complete report detailing the results of all testing performed. The report shall list the serial numbers of equipment tested during the reporting period. During the conduct of tests performance parameters shall be measured and verified in accordance with this specification. The report shall include completed Form NAVEXOS 3373 for the shock test. Equipment will not be accepted unless the report is complete.

Source: https://assist.dla.mil -- Downloaded: 202610-20T13:49Z
Check the source to verify that this is the current version before use.

4.5 Inspection procedures.-

4.5.1 Surface examination.- Equipment shall be examined for the following:

(a) Workmanship, assembly and fit, mechanical safety, and marking
(b) Materials, parts, coatings, and finish

4.5.2 Operating test.- The equipment shall be energized completely aligned and calibrated and subjected to an operating test to insure qualitatively the proper functioning of the equipment, including all operating controls and conformance with the safety requirements. This test shall include supply line and frequency tests, overload protection tests, output voltage and current, ripple, regulation and parallel operation to verify compliance with the applicable detailed requirements specified herein.

4.5.3 All other inspections referenced herein shall be conducted in accordance with the Quality Assurance Provisions of MIL-E-16400, as applicable.

4.5.4 Inspection of preparation for delivery.- The inspection of the preservation and packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing and marking.- Unless otherwise specified, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing and marking specified in MIL-E-17555 (see 6.2).

6 NOTES

6.1 Intended use.- The equipment covered by this specification is intended for service in ship and shore stations where it is expected to withstand continuous use for long periods, under military service conditions, without benefit of overhaul. The equipment is in each case a vital instrument; failure at a critical moment invariably results in serious reduction in the battle efficiency. Emergency repairs can seldom be made with sufficient celerity to avoid having equipment out of service during a critical period.

6.2 Ordering data.- Procurement documents should specify the following:

6.2.1 Procurement requirements:

(a) Title, number, and date of this specification.
(b) Number of first article samples if other than specified in 4.3.
(c) Selection of applicable levels of preservation, packaging, packing, and marking (see 5.1).

Source: https://assist.dla.mil -- Downloaded: 202610-20T13:49Z
Check the source to verify that this is the current version before use.

6.2.2 Contract data requirements.- Data generated by this document is not deliverable unless specified on the Contract Data Requirements List (DD Form 1423). The data required by this specification is as follows:

(a) Reliability program plan (see 3.6.2)
(b) Reliability prediction (see 3.6.3)
(c) Reliability reports (see 3.6.5)
(d) Instruction manuals if required (see 3.8)
(e) Inspection and quality control plan (see 4.1.1)
(f) Failure reporting in accordance with MIL-E-16400 (see 4.1.3)
(g) Test, test equipment, and test procedures (see 4.1.4)
(h) First article inspection report (see 4.3.2)
(i) Nonconfromance test report (see 4.4.4)
(j) Quality conformance inspection report (see 4.4.6)

6.3 First article.- Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Preparing activity:
Navy - EC
(Project 5815-N059)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:49Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

Source: https://assist.dla.mil -- Downloaded: 20260410-20T13:49Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:49Z
Check the source to verify that this is the current version before use.